18. (d) – 635.66 kJ

19. (c) 80 kJ mol^{-1}

20. (a) $\Delta H < 0$ and $\Delta S > 0$

21. (c) adiabatic expansion

22. (d) (–, –, +)

23. (b) 27 ^{0}C

24. (d) $CaCO_3(S) \rightarrow CaO(S) + CO_2(g)$

25. (a) 300K

Keys to multiple choice questions:

7.

$w = - P \Delta V$

$w = - (1 \times 10^5 \text{ Nm}^{-2})$

$(1 \times 10^{-2} \text{ m}^3 - 1 \times 10^{-3} \text{ m}^3)$

$w = -10^5 (10^{-2} - 10^{-3})$ Nm

$w = - 10^5 (10 - 1) 10^{-3})$ J

$w = - 10^5 (9 \times 10^{-3})$ J

$w = -9 \times 10^2$ J

$w = -900$ J

9.

$$CO(g) + \frac{1}{2} O_2(g) \rightarrow CO_2(g)$$

$\Delta H_C^0 (CO) = \Delta H_f (CO_2) - \Delta H_f(CO) + \Delta H_f (O_2)]$

$\Delta H_C^0 (CO) = - 94$ KCal $- [- 26.4$ KCal $+ 0]$

$\Delta H_C^0 (CO) = - 94$ KCal $+ 26.4$ KCal

$\Delta H_C^0 (CO) = - 67.6$ KCal

11.

$2Al + Cr_2O_3 \rightarrow 2Cr + Al_2O_3$

$\Delta H_r^0 = [2\Delta H_f (Cr) + \Delta H_f (Al_2O_3)]$

$- [2\Delta H_f (Al) + \Delta H_f (Cr_2O_3)]$

$\Delta H_r^0 = [0 + (- 1596 \text{ kJ})]$

$- [0 + (- 1134)]$

$\Delta H_r^0 = - 1596$ kJ $+ 1134$ kJ

$\Delta H_r^0 = - 462$ kJ

14.

$\Delta U = q + w$

$\Delta U = - 1$ kJ $+ 4$ kJ

$\Delta U = + 3$kJ

15.

$Fe + 2HCl \rightarrow FeCl_2 + H_2$

1 mole of Iron liberates 1 mole of Hydrogen gas

55.85 g Iron = 1 mole Iron

$\therefore n = 1$

$T = 25^0$ C = 298 K

$w = - P \Delta V$

$$w = - P \left(\frac{nRT}{P} \right)$$

$w = - nRT$

$w = -1 \times 8.314 \times 298$ J

$w = - 2477.57$ J

$w = - 2.48$ kJ

16.

$T_i = 125^0$ C = 398 K

$T_f = 25^0$ C = 298 K

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